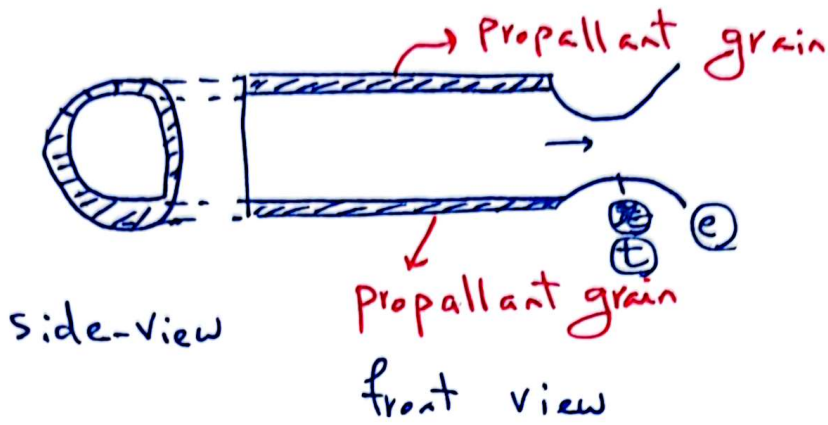


Solid Rocket Motors, ch. 12 (Sec. 4)



Solid propellants

Homogenous

Fuel and oxidizer in the same molecules

Ex: Double-base propellant (nitroglycerin-nitrocellulose)

Heterogenous

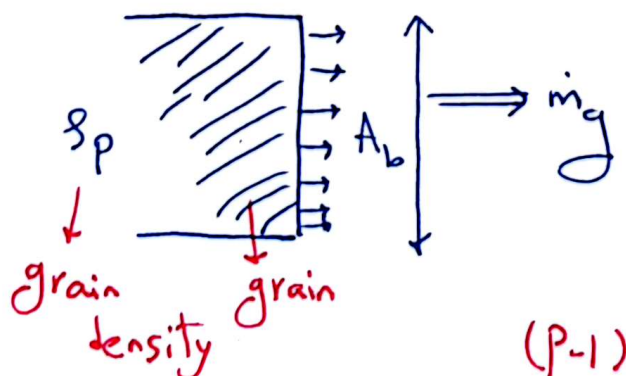
Fuel and oxidizer are mixed.

Ex: Composite grain
① plastic-like binder: HTPB

② oxidizing crystals: AP, AN, NP, KP, KN

③ fuel metal additives: Al, Mg, B

Solid propellants are observed to regress at a uniform rate during their burn (regression rate r)



$$\dot{m}_g = s_p A_b r$$

The regression rate is related to pressure as follows:

$$r = \underline{a} p_o^n \quad (12.25)$$

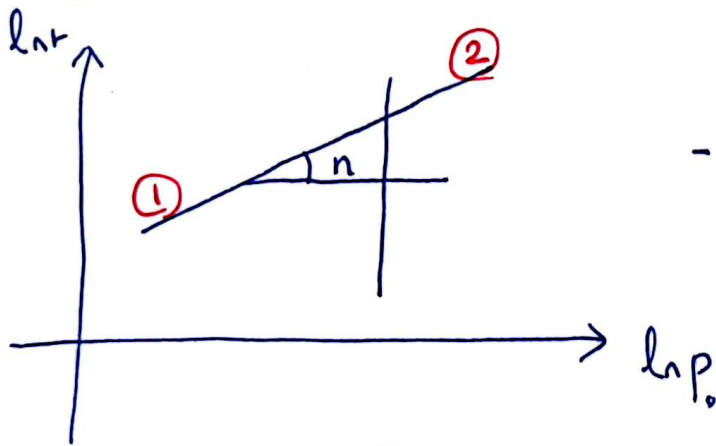
n : pressure exponent (independent of temperature)

a : depends on initial temperature of the grain (T_p)

$$(12.26) \quad a = \frac{A}{T_i - T_p} \quad (A \text{ and } T_i \text{ are constants})$$

$$\text{From (12.25)} \Rightarrow \ln r = \ln a + n \ln p_o$$

To get n , we need two r values and two p_o values.



$$\begin{aligned} \ln r_2 &= \ln a + n \ln p_{o2} \\ - \ln r_1 &= \ln a + n \ln p_{o1} \\ \hline \ln(r_2) - \ln(r_1) &= n(\ln p_{o2} - \ln p_{o1}) \end{aligned}$$

$$\text{or } n = \frac{\ln\left(\frac{r_2}{r_1}\right)}{\ln\left(\frac{p_{o2}}{p_{o1}}\right)}$$



$$\dot{m}_g = \dot{m}_t$$

$$\dot{m}_g = \rho_p A_b r = \rho_p A_b a p_o^n$$

$$\dot{m}_t = \frac{p_o}{\sqrt{RT_o}} \sqrt{\gamma} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}} A_t$$

Put $\dot{m}_g = \dot{m}_t \Rightarrow p_o = \left[\frac{A_b}{A_t} \frac{a \rho_p}{\sqrt{\frac{\gamma}{RT_o}} \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma+1}{\gamma-1}}} \right]^{\frac{1}{1-n}} \quad (12-29)$

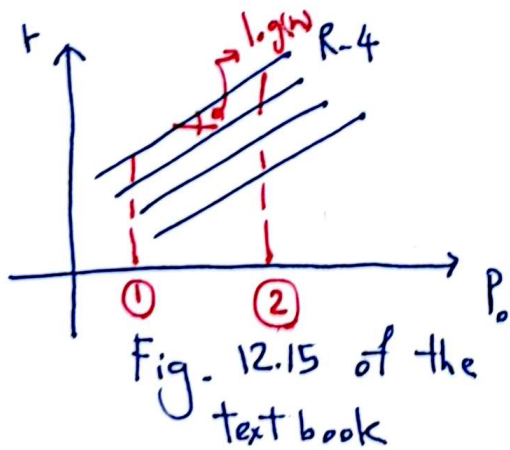
$$\frac{1}{c^*} \quad (11-9)$$

$$p_o = \left(\frac{A_b}{A_t} a \rho_p c^* \right)^{\frac{1}{1-n}}$$

Example: Find $\left(\frac{A_b}{A_t} \right)$ required to reach a steady-state operating pressure of 14 MPa in a solid rocket motor with the following characteristics:

$$\rho_p = 1710 \frac{\text{kg}}{\text{m}^3}, \quad \gamma = 1.27, \quad T_o = 2220 \text{ K}, \quad \bar{M} = 23$$

Propellant is R-4, see Fig. 12-15 for burn data.



Solution: From this graph:

$$\begin{cases} \text{At } p_{o1} = 1.0 \text{ atm}, r_1 = 0.07 \frac{\text{cm}}{\text{s}} \\ \text{At } p_{o2} = 10.0 \text{ atm}, r_2 = 0.22 \frac{\text{cm}}{\text{s}} \end{cases}$$

$$\Rightarrow n = \frac{\ln \left(\frac{r_2}{r_1} \right)}{\ln \left(\frac{p_{o2}}{p_{o1}} \right)} = \frac{\ln \left(\frac{0.22}{0.07} \right)}{\ln \left(\frac{10.0}{1.0} \right)} \Rightarrow$$

$$n = 0.4973$$

(12-3)

First, we got n , then from one point data, get "a"

$$a = \frac{r_1}{P_{o1}^n} = \frac{0.07 \times 10^{-2}}{[10 \times 1.1325]^{0.4973}} = 7.2186 \times 10^{-7} \left(\frac{\frac{m}{s}}{P_a^{0.4973}} \right)$$

$$P_o = 14 \text{ MPa}, \quad \gamma_p = 1710 \frac{\text{kg}}{\text{m}^3}, \quad \gamma = 1.27, \quad R = \frac{\bar{R}}{M} = 361.5 \frac{\text{J}}{\text{kgK}}$$

$$C^* = \sqrt{\frac{1}{\gamma} R T_o \left(\frac{\gamma+1}{2} \right)^{\frac{\gamma+1}{\gamma-1}}} = 1353.7 \frac{\text{m}}{\text{s}}$$

$$P_o = \left(\frac{A_b}{A_t} a \gamma_p C^* \right)^{\frac{1}{1-n}} \Rightarrow \frac{A_b}{A_t} = \frac{P_o^{1-n}}{a \gamma_p C^*} = 2341$$

$$r = a P_o^n = 7.2186 \times 10^{-7} \times (14 \times 10^6)^{0.4973} = 0.002584 \frac{\text{m}}{\text{s}} = 0.258 \frac{\text{cm}}{\text{s}}$$

See Fig. 12.17 for schematic of several internal burning grain and associated thrust variation.

One of the configurations of propellant grain is called end-burning

